

VALENCIA AP, Spain

WMO: 082840

Lat: 39.485N

Lon: 0.475W

Elev: 203

StdP: 14.59

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	33.4	35.8	15.8	12.4	46.2	20.8	15.7	48.2	31.6	60.0	27.8	58.3	5.3	280	0.413

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.9	92.7	69.3	89.7	70.4	87.7	70.9	76.6	84.4	75.4	83.4	74.4	82.5	11.2	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.9	127.5	81.3	73.0	123.6	80.9	71.6	117.7	80.0	40.1	84.4	39.0	83.5	38.0	82.5	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.2	20.7	17.7	28.4	101.4	3.0	4.3	26.2	104.5	24.4	107.0	22.7	109.4	20.5	112.5		
(5)			25.7	77.9	3.3	2.5	23.3	79.7	21.4	81.2	19.5	82.6	17.1	84.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.0	51.4	52.4	56.3	60.3	66.3	73.5	78.4	79.0	73.4	66.8	57.6	52.3		
	DBStd	10.95	5.46	5.44	5.32	4.49	4.66	4.18	2.95	3.12	4.28	4.79	5.60	5.30		
	HDD50	130	45	32	8	1	0	0	0	0	0	0	7	37		
	HDD65	1904	423	354	275	153	38	1	0	0	1	35	229	395		
	CDD50	5255	88	99	204	308	506	705	880	900	701	522	235	107		
	CDD65	1555	0	1	5	11	79	256	415	435	253	92	7	1		
	CDH74	13758	2	7	63	108	564	2146	4125	4371	1824	510	37	1		
	CDH80	4558	0	0	8	12	126	649	1513	1660	490	95	5	0		
(14) Wind	WSAvg	7.2	8.0	8.0	8.0	7.9	7.0	6.8	6.9	6.7	6.6	6.1	7.2	7.2		
(15) Precipitation	PrecAvg	17.7	1.3	1.2	1.4	1.2	1.3	0.9	0.2	0.6	2.7	3.1	2.0	1.7		
	PrecMax	32.8	6.0	4.2	4.6	3.5	5.6	5.2	1.3	2.0	10.2	14.4	8.0	6.6		
	PrecMin	8.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	6.5	1.4	1.1	1.3	0.9	1.5	1.1	0.3	0.5	2.7	3.9	2.2	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.1	74.8	80.4	81.2	88.3	94.8	95.7	98.7	93.0	87.0	78.6	71.2		
		MCWB	55.6	56.1	59.1	60.4	63.5	68.5	67.4	68.7	67.6	65.2	60.8	57.2		
	2%	DB	68.0	69.7	74.9	77.1	83.0	89.4	91.4	92.6	87.7	82.3	73.6	67.7		
		MCWB	53.9	54.4	56.9	58.4	63.2	67.2	70.4	70.4	68.8	64.6	58.8	55.5		
	5%	DB	64.6	66.3	71.1	73.6	79.3	86.2	89.0	89.6	84.9	79.1	70.2	64.8		
		MCWB	52.4	52.8	54.9	57.0	62.0	67.9	71.1	72.1	69.3	64.7	57.8	53.9		
	10%	DB	61.9	63.0	67.6	70.9	76.7	83.6	87.0	87.6	82.6	76.6	67.5	62.4		
		MCWB	51.3	51.3	53.9	56.8	61.6	67.3	71.5	72.5	69.0	64.3	56.6	52.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.9	58.3	61.4	63.2	68.8	75.2	77.4	78.3	76.2	71.0	64.4	59.5		
		MCDB	68.0	70.3	75.0	74.1	79.3	83.4	85.5	86.1	83.0	77.0	72.2	67.6		
	2%	WB	55.7	56.1	58.5	61.1	66.6	72.7	76.0	77.1	74.2	69.2	62.0	57.4		
		MCDB	64.3	66.0	70.1	71.0	76.4	81.8	84.2	85.0	81.3	76.1	69.2	64.6		
	5%	WB	54.1	54.3	56.9	59.6	65.1	71.1	74.8	76.1	72.7	67.6	60.2	55.5		
		MCDB	61.9	63.1	67.0	69.2	74.5	80.5	83.3	83.9	80.1	74.8	66.9	62.7		
	10%	WB	52.5	52.8	55.6	58.3	63.9	69.7	73.8	75.0	71.4	66.2	58.6	53.9		
		MCDB	60.0	60.6	64.2	67.6	73.0	79.2	82.5	82.9	79.1	73.6	65.4	60.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.2	19.5	19.9	19.1	18.8	18.3	17.1	16.9	17.3	17.8	18.2	18.7		
		MCDBR	22.8	24.5	25.9	24.4	23.2	22.5	20.9	20.9	20.7	22.1	21.4	21.3		
		MCWBR	12.6	13.0	12.5	11.1	9.6	9.1	9.1	9.6	9.4	10.3	11.3	12.0		
	5% WB	MCDBR	18.9	21.2	21.7	20.8	18.9	18.0	17.1	16.5	16.9	18.1	17.9	18.4		
		MCWBR	11.6	12.4	12.0	10.8	9.0	8.5	8.5	8.7	8.7	9.6	10.5	11.3		
(40) Clear-Sky Solar Irradiance	taub		0.306	0.328	0.373	0.391	0.417	0.437	0.435	0.441	0.419	0.379	0.334	0.311		
	taud		2.521	2.479	2.370	2.336	2.294	2.277	2.299	2.286	2.339	2.442	2.514	2.526		
	Ebn at Noon		276	284	279	281	274	268	267	262	261	261	262	265		
	Edh at Noon		24	29	35	39	42	42	41	41	36	30	24	22		
(44) All-Sky Solar Radiation	RadAvg		749	1007	1396	1744	2063	2359	2343	2003	1541	1125	791	646		
	RadStd		85	114	137	131	148	92	77	90	96	100	89	56		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	-4.14	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	-2.43	+0.74	N/A	N/A	N/A	N/A	N/A	+88

Nomenclature: See separate page